



Representing and Interpreting Data

Worksheet 1

Statistics Chapter 3: Representing and Interpreting Data. Answer all questions. Show your working clearly. Total: 65 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- On a histogram, **area** represents frequency. Comparisons must cover location **and** spread, **in context**.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Histograms	20	
B Cumulative frequency diagrams and box plots	21	
C Comparing distributions	19	
Roar-level challenge	5	
Total	65	

Section A Histograms

20 marks

1. Find the frequency density of (a) the class $10 \leq x < 20$ with frequency 30, (b) the class $20 \leq x < 25$ with frequency 20. [2]

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2. Calculate the frequency density for each class in the table. [3]

Length, l (cm)	$0 \leq l < 10$	$10 \leq l < 15$	$15 \leq l < 20$	$20 \leq l < 30$	$30 \leq l < 50$
Frequency	12	18	25	20	10
Frequency density					



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3. On a histogram with frequency density on the vertical axis, the bar for the class $20 \leq x < 30$ has height 4.5 and the bar for the class $30 \leq x < 35$ has height 2. **[3]**
Find the frequency of each of these two classes.

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4. On a histogram, the bar for the class $10 \leq t < 15$ is 2 cm wide and 6 cm tall, and represents 24 people. **[4]**
The bar for the class $15 \leq t < 25$ is 4 cm wide and 2.5 cm tall.
(a) Find the number of people represented by 1 cm^2 .
(b) Find the frequency of the class $15 \leq t < 25$.

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5. The times, t minutes, taken by some customers to be served are summarised below.

Time, t (min)	$0 \leq t < 10$	$10 \leq t < 15$	$15 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$
Frequency	10		30	24	6

On a histogram of these data, the bar for $15 \leq t < 20$ is 1 cm wide and 6 cm tall, and the bar for $10 \leq t < 15$ is 5 cm tall.

(a) Find the frequency of the class $10 \leq t < 15$. [3]

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(b) Find the width and height of the bar for the class $20 \leq t < 30$. [2]

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(c) Estimate the number of customers who took between 12 and 25 minutes to be served. [3]

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(Total for Question 5 is 8 marks)



Section B Cumulative frequency diagrams and box plots

21 marks

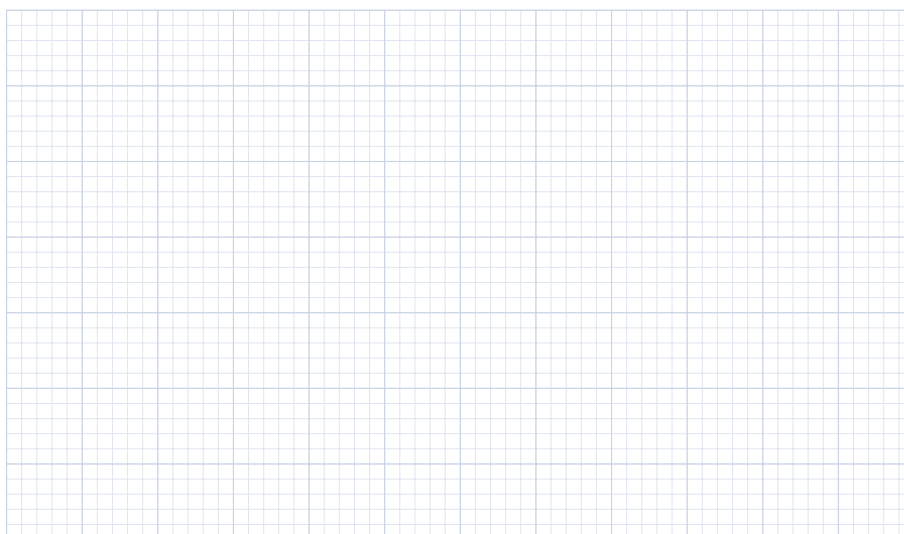
6. Complete the cumulative frequency column for the ages of 50 people.

[2]

Age, a (years)	Frequency	Cumulative frequency
$0 \leq a < 10$	5	
$10 \leq a < 20$	12	
$20 \leq a < 30$	18	
$30 \leq a < 40$	10	
$40 \leq a < 50$	5	

7. Using your answer to Question 6, draw a cumulative frequency diagram on the grid below (age on the horizontal axis, cumulative frequency on the vertical axis), and use it to estimate the median age.

[3]





8. Using the table in Question 6 and linear interpolation, estimate the lower quartile, the upper quartile and the interquartile range of the ages. [4]

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9. The times, in minutes, of a group of runners have minimum 12, $Q_1 = 18$, median 23, $Q_3 = 27$ and maximum 45. [4]

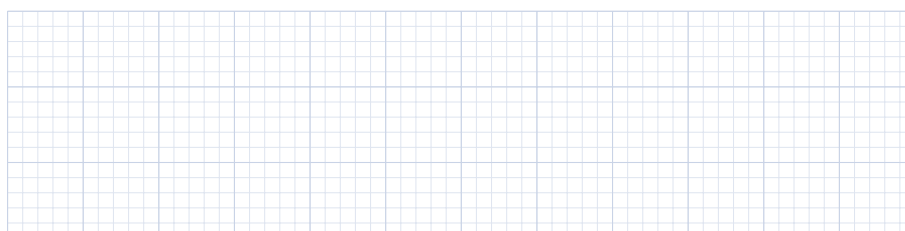
(a) Using the $1.5 \times \text{IQR}$ rule, show that 45 is an outlier.

(b) The largest time that is not an outlier is 38. Draw a box plot of the data on the grid below.

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10. The marks of 60 students in a test are summarised below.

Mark, m	$0 \leq m < 20$	$20 \leq m < 40$	$40 \leq m < 60$	$60 \leq m < 80$	$80 \leq m < 100$
Frequency	4	10	22	18	6

(a) Find the cumulative frequencies.

[2]

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(b) Use linear interpolation to estimate the median and the interquartile range.

[4]

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(c) The teacher wants 75% of the students to pass. Estimate the pass mark she should set.

[2]

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(Total for Question 10 is 8 marks)



Section C Comparing distributions

19 marks

11. The test marks of Class A have median 62 and IQR 15. The test marks of Class B have median 58 and IQR 8. [2]
Compare the marks of the two classes.

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12. In one month, Town P had mean daily rainfall 2.1 mm with standard deviation 3.4 mm, and Town Q had mean daily rainfall 1.4 mm with standard deviation 1.2 mm. [3]
Compare the rainfall in the two towns, and explain what the large standard deviation for Town P suggests.

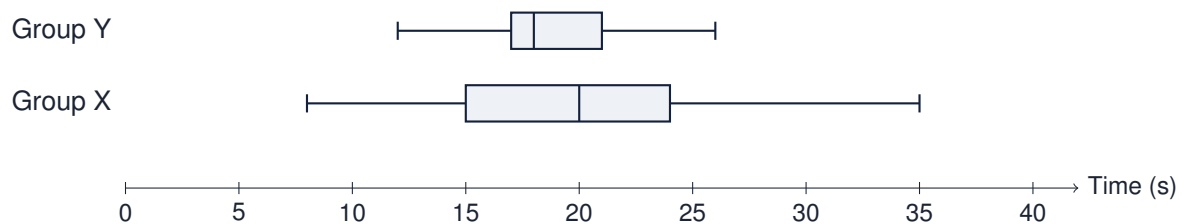
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13. The box plots show the times, in seconds, taken by two groups of students to complete a puzzle. [3]



Compare the times of the two groups.

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14. Explain why the median and interquartile range are usually better than the mean and standard deviation for comparing house prices in two towns. [3]

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15. The daily maximum temperatures, x °C, at Leuchars and Hurn in May 2015 are summarised below (large data set).

	n	Σx	Σx^2
Leuchars	31	402.6	5357.2
Hurn	31	520.8	8872.4

- (a) Calculate the mean and standard deviation of the daily maximum temperature at each location. [4]

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- (b) Compare the daily maximum temperatures at the two locations. [2]

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- (c) Leuchars is in Scotland and Hurn is in the south of England. Comment on whether your comparison is what you would expect, and on whether it applies to the whole year. [2]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A histogram for the data below has bars with frequency densities 2.4, k and 1.2 for the classes $0 \leq x < 10$, $10 \leq x < 15$ and $15 \leq x < 30$. The total frequency is 80. Find k , and use interpolation to estimate the median. [5]

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END OF WORKSHEET

Total: 65 marks